

REDUCED-ORDER MODELING OF THE RUL OF LI-ION BATTERIES

MMAE 500 DATA DRIVEN MODELING

By Zaid Khan | 08/05/2026

Abstract

Accurate prediction of lithium-ion battery Remaining Useful Life (RUL) is critical for optimizing electric vehicle thermal management systems and ensuring operational safety. This study initially aimed to construct a multi-parameter Reduced-Order Model (ROM) to simulate voltage degradation profiles across continuous operational domains, such as varying discharge rates and ambient temperatures. High-level data extraction, rigorous filtering of characterization pulses, and Depth-of-Discharge (DoD) normalization were executed on the CALCE and NASA empirical battery datasets to construct standardized spatial snapshot matrices. However, an analysis of these datasets revealed a critical limitation: erratic degradation behaviors, including transient capacity regeneration and intersecting thermal degradation paths, violated the monotonic correlation and orthogonality constraints required for spatial manifold interpolation, rendering multi-parameter modeling unviable.

Consequently, the methodology pivoted to a single-parameter temporal ROM utilizing Singular Value Decomposition (SVD). While SVD spatial bases frequently serve as foundational feature extractors for more advanced Scientific Machine Learning (SciML) and hybrid data-driven architectures, in this framework, SVD was explicitly utilized to isolate the underlying linear dynamics of the capacity fade into a truncated orthogonal basis. To reconstruct the battery lifecycle without requiring continuous physical testing, the temporal coefficients were interpolated utilizing exclusively two physical boundary profiles: Beginning of Life (BOL) and End of Life (EOL). A comparative analytical evaluation demonstrated that Gaussian interpolation significantly outperformed standard linear interpolation; for instance, at prediction cycle 141, the error using Gaussian interpolation was 0.00789, nearly halving the linear interpolation error of 0.015. Ultimately, while empirical dataset constraints restricted multi-parameter scalability, the proposed Gaussian-SVD framework establishes a highly accurate, computationally efficient surrogate model for battery prognostics that relies on minimal experimental training data.

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1. Motivation

The design of robust enclosures for Electric Vehicle (EV) battery packs is strictly governed by rigorous thermal and safety constraints, particularly those outlined by ASME and European regulatory standards to mitigate thermal runaway. A critical extension of enclosure thermal management is the accurate prognostics of the lithium-ion batteries themselves, specifically, modeling the Remaining Useful Life (RUL) under varying operational and environmental conditions. Understanding how extreme thermal loads and discharge rates influence capacity fade is essential for optimizing both battery efficiency and systemic safety.

A primary challenge in RUL prognostics is predicting degradation behavior across continuous operational domains using limited, discrete experimental data. For instance, thermal management systems require accurate RUL estimations at intermediate ambient temperatures (e.g., 32°C), necessitating the interpolation of physical boundary data recorded at 24°C and 44°C. Singular Value Decomposition (SVD) is frequently utilized to extract these underlying physical structures from experimental data, capturing the linear dynamics of degradation through discrete spatial modes. When properly formulated, these modal bases align with known governing electrochemical equations and validate theoretical degradation models.

However, the fidelity of any data-driven model is inherently constrained by the physical reality of the experimental data. Battery degradation kinetics are highly complex and frequently non-monotonic; phenomena such as erratic capacity regeneration or accelerated solid-electrolyte interphase (SEI) layer growth at elevated temperatures can severely skew linear projections. Consequently, applying a singular, global SVD-basis across disparate physical conditions is often mathematically insufficient and can lead to structurally invalid interpolations.

This paper investigates the limitations of utilizing a global spatial basis for battery prognostics and demonstrates the necessity of hybrid modeling to capture these localized, non-linear dynamics. By evaluating the constraints of traditional linear interpolation on highly variable experimental data, this study will explore the transition toward more advanced, robust modeling techniques capable of accurately mapping multi-parameter degradation spaces.

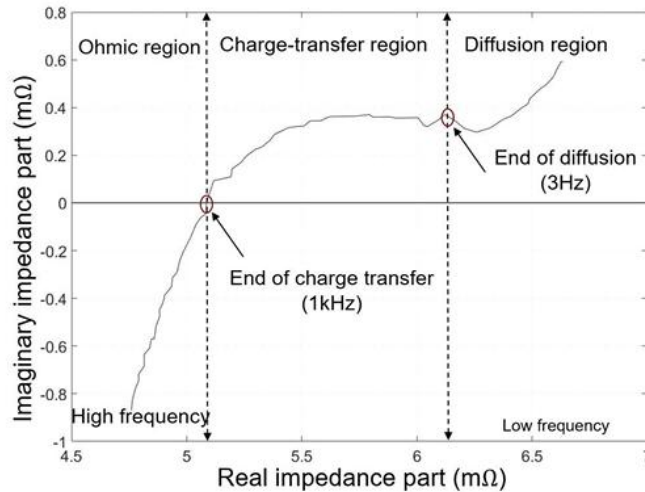
2. Background Research

The degradation of battery health is a central point of concern for sustainability-driven endeavors. Batteries, like Li-Ion used in the automotive industry, offer an eco-friendly alternative energy source. It becomes imperative to predict the SOH (State of Health) and model RUL (Remaining Useful Life) to ascertain degradation trajectory (Elmahallawy et al., 2022). Due to the abundance of factors causing adverse impact to the battery's health, modeling for all scenarios becomes nearly impossible. The main goal is to characterize the aging process, which constitutes analysis of parameter variation from BOL (Beginning of Life) to EOL (End of Life). The capacity of the battery is the amount of energy stored in one cell. The SOH can be described as –

$$SOH = \text{ratio of capacity at } \left(\frac{BOL}{EOL} \right)$$

The measure of how the capacity fades over time is done through parametric analysis of electrical/thermal variables – voltage (V), current (A) and temperature (T). The charging and discharging cycles accelerate degradation. The power capacity of the battery is measured via coulomb counting. Another parameter of interest is the electrical impedance, measured via EIS (Electrochemical impedance spectroscopy).

Figure 2-1; Nyquist plot of EIS spectrum for Li-ion battery internal impedance.



2.1. Error Metrics

1. The Transient Reconstruction Error (The Full Matrix) –

To measure how well you captured the entire voltage/temperature discharge curve, use the Relative Frobenius Norm.

$$E_{recon} = \frac{||X_{true} - \hat{X}||_F}{||X_{true}||_F}$$

Where $|| \cdot ||_F$ is the square root of the sum of the absolute squares of its elements. This yields a percentage error (e.g., 0.05 means a 5% error across the whole physical field).

2. The Target Metric (Capacity/RUL) –

We must extract the predicted Capacity from your reconstructed matrix and compare it to the true Capacity using Root Mean Squared Error (RMSE).

$$RMSE_{cap} = \sqrt{\frac{1}{N} \sum_{i=1}^N (C_{true}^{(i)} - \hat{C}^{(i)})^2}$$

2.2. Datasets

While the performance of the battery is entirely dependent on its chemistry and operating conditions, we aim to isolate one variable. NASA-Ames and CALCE provide datasets of the impedance, charge, and RUL of many Li-Ion batteries.

The dataset was accessed via Kaggle, where many have leveraged various modeling techniques to predict RUL or extract interesting characteristics from the data.

URL: [NASA Battery Dataset](#)

Other sources:

- CALCE Battery Research Group (University of Maryland)
- Sandia National Laboratories (EV)

2.3. Aims

This primary objectives of this paper are to –

- Extract the high energy modes from reconstructed global SVD that can be used to predict RUL characteristics for Li-Ion batteries using NASA dataset.
- Perform multi-parameter modeling on Li-Ion (CS2) batteries from the CALCE dataset.

3. Data Preprocessing & Exploratory Analysis

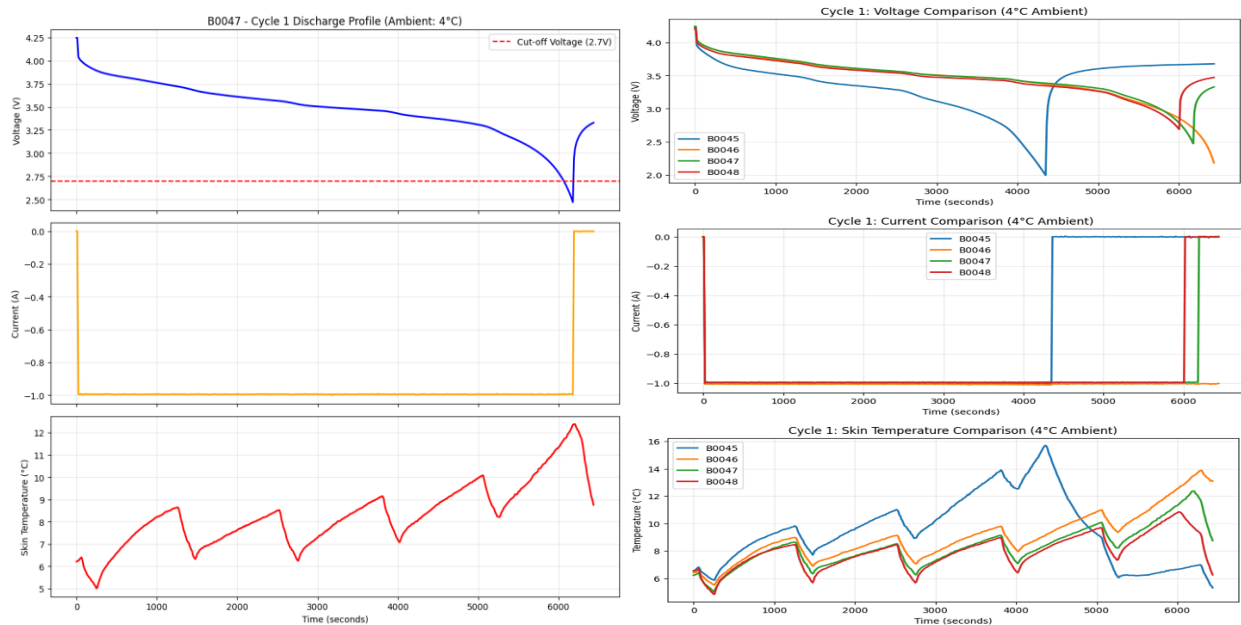
Before diving into modeling the data, we must first understand the operating conditions of the experiment that it was sourced from.

3.1. Dataset Selection and Operational Profile Isolation

The integrity of a multi-parameter subspace model relies heavily on the quality and structure of the input data. To develop a robust predictive model for Remaining Useful Life (RUL) that can ultimately inform the thermal constraints of an EV enclosure, it is necessary to isolate datasets that exhibit consistent degradation mechanics.

The NASA Ames Prognostics Data Repository contains lithium-ion cells subjected to various environmental profiles. An initial exploratory analysis was conducted to compare cells operated under strictly isothermal conditions (e.g., a constant ambient 24°C or 43°C) against those subjected to dynamic thermal profiles (e.g., alternating between 24°C and 44°C).

Figure 3-1: (a) Left plots represent voltage, current and temperature profiles during discharge for B0047, (b) right plots provides a comparison between the profiles of the same battery group.



To develop a robust model, we isolate the datasets with unchanging ambient temperatures to avoid discontinuities in the profiles. In B0040, there is a change in ambient temperature at the 15th discharge cycle in concurrence with a 240% increase in capacity from 0.715 to 1.708 Ahr as seen in Figure 3-1. We want to avoid these parameter differences when modeling. When observing the capacity degradation for the B0034, which ran for much longer cycles ~195, we see small hump like structures and some huge spikes.

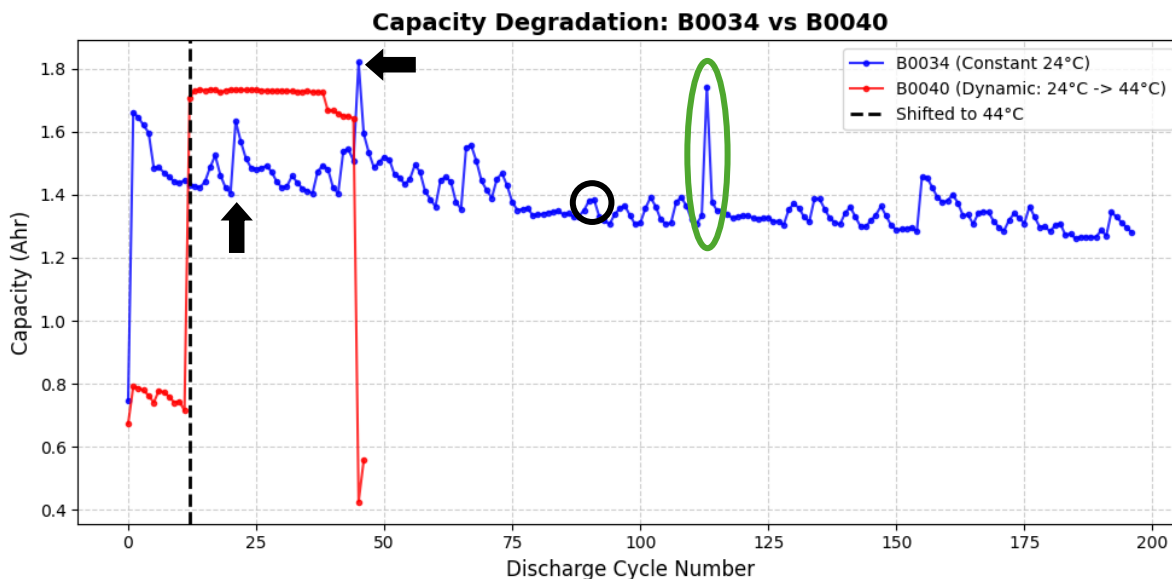
Metric of Comparison – The primary parameter utilized to evaluate the stability of the degradation trajectory was the measured Discharge Capacity (Ahr) over the cell's cycle life. Capacity serves as the definitive indicator of overall cell health and is the target metric for RUL prediction.

Cells operated under isothermal conditions, specifically the 24°C group (B0005, B0006, B0007, B0018) and the 43°C group (B0029, B0030, B0031, B0032), exhibit continuous, monotonic capacity fade.

In analyzing the baseline isothermal degradation trajectories (e.g., Battery B0034 in Figure 3-2), two distinct types of data variance were observed: minor, localized capacity recoveries and severe, instantaneous capacity spikes. To construct a highly accurate transient data matrix for Subspace Modelling, distinguishing between real physical phenomena and hardware artifacts is a critical preprocessing requirement.

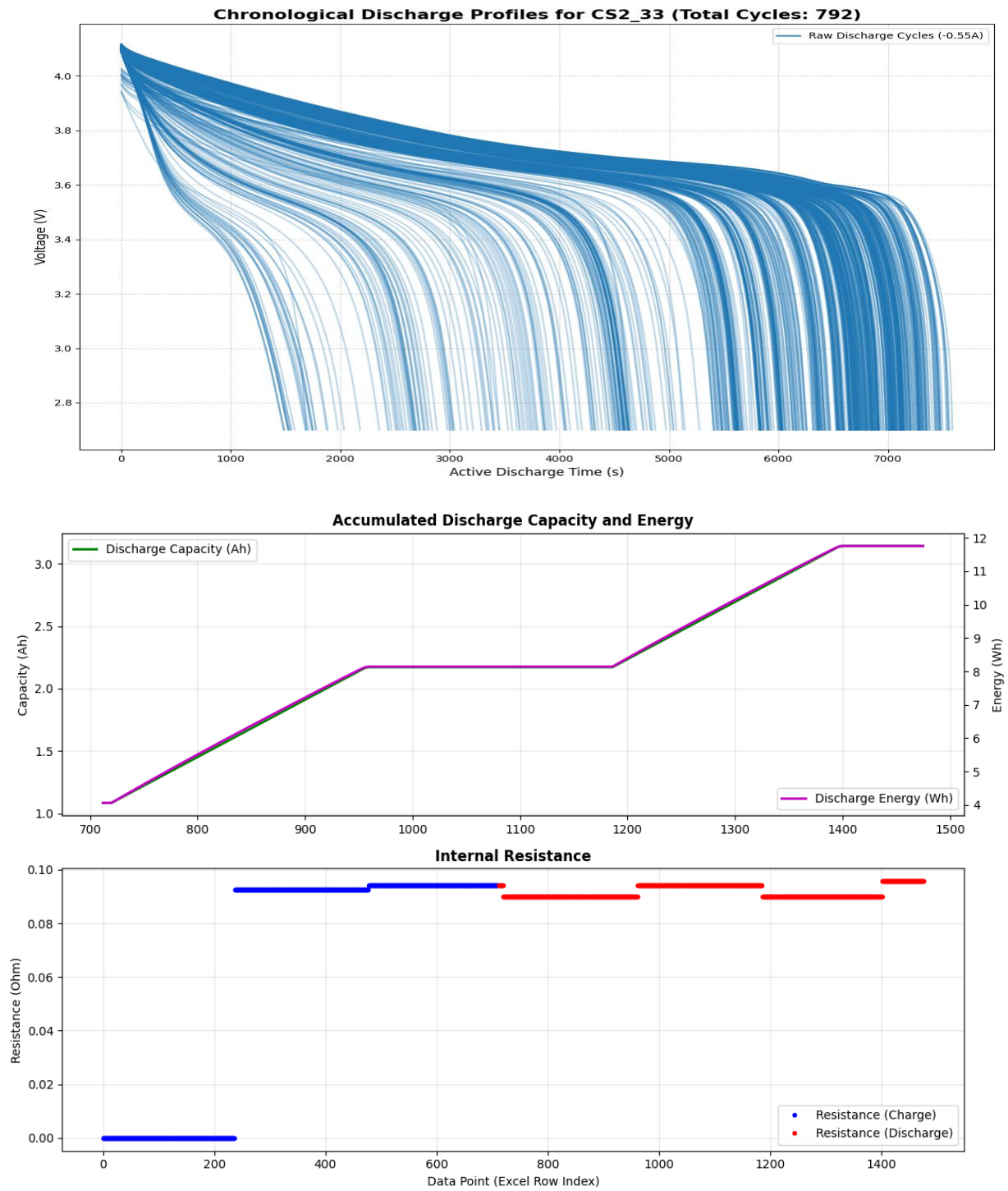
Between the 19th and 20th cycle (represented by black arrow in Figure 3-2), there was a three-day gap. That explains the massive discontinuity seen in the plot. Similarly, in cycle 46, the data shows a 13-day gap. However, the big spike in green only had a 9-hour gap.

Figure 3-2: Compares the degradation of battery capacity (Ahr) per cycle between B0034 (isothermal) and B0040 (dynamic) dataset. The black circle points to the minor fluctuations and green ellipse to the spikes.



CALCE Dataset (CS2 Battery)

Figure 3-3: Voltage, current, discharge energy, and internal resistance variation in one charging and discharging cycle of CS2_33 battery.

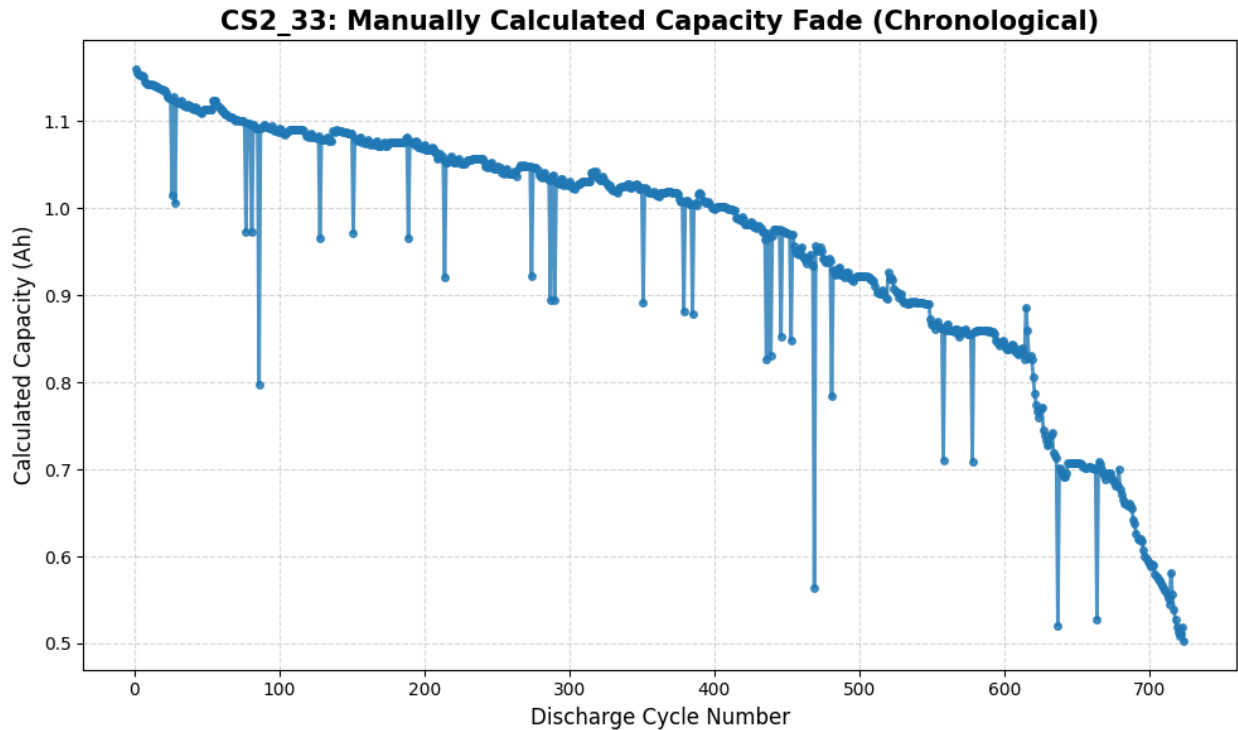


The data comprises of commercial lithium-ion cells subjected to rigorous life cycle testing to characterize parameter-dependent degradation. The test cells possess a nominal capacity of 1.1 Ah and were cycled within a standard operating voltage window (typically 4.2 V to 2.7 V). To isolate the effects of varying operational loads, the batteries were grouped and discharged at distinct constant-current rates (C-rates) at room temperature. Specifically, the CS2_33 group underwent a 0.5C (0.55 A) continuous discharge, while the CS2_35 group was subjected to a 1.0C (1.1 A) load. All cells were recharged using a standard Constant Current-Constant Voltage (CC-CV) protocol. This controlled parametric variance provides the necessary boundary conditions to formulate a robust, multi-parameter Reduced-Order Model (ROM) mapping stress-induced capacity fade.

We isolate the discharging cycles from the dataset (-0.55A constant current applied). We see that the discharge capacity and energy values rise during discharging. Like NASA dataset, the discharge cycles tend to grow shorter and fall off sooner as the number of cycles increases. We derive the capacity fade manually through –

$$Q = \int I dt$$

Figure 3-4: Capacity fade of CS2_33.



3.1.1. Physical Capacity Regeneration (The Relaxation Effect)

The minor fluctuations can be attributed to a well-known electrochemical phenomenon known as the "relaxation effect." During extended rest periods between active cycling,

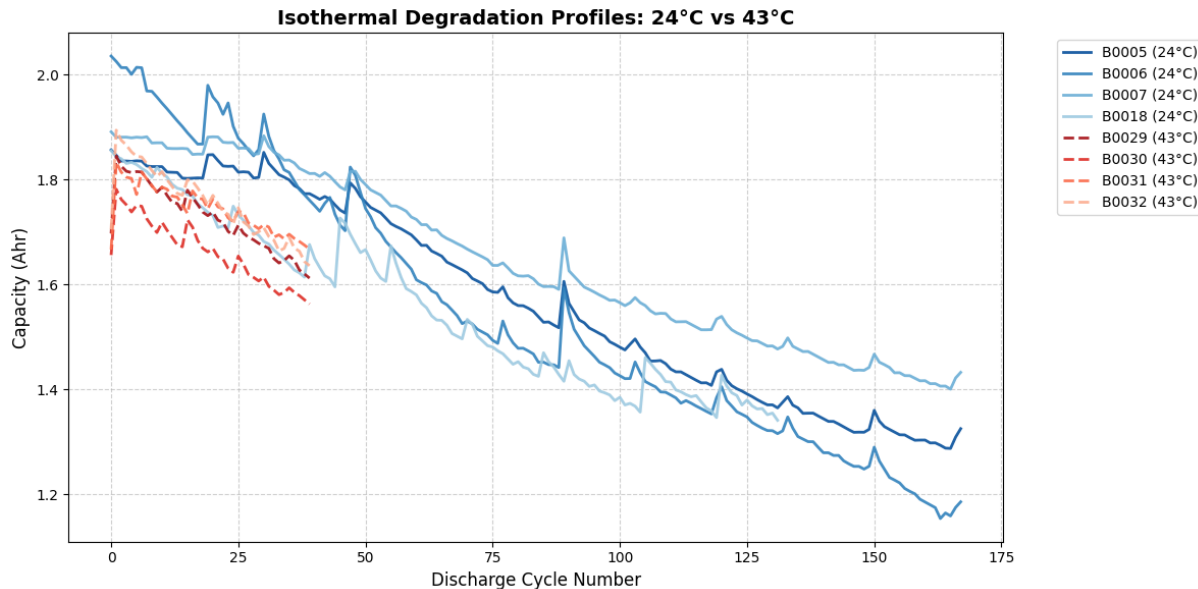
lithium ions that have unevenly aggregated at the electrodes are allowed time to diffuse and establish a more uniform concentration gradient (Sun et al., 2024). This temporary equilibration slightly reduces the cell's internal resistance, yielding a marginal, transient recovery in discharge capacity on subsequent cycles before the overarching degradation trend resumes.

3.1.2. Hardware-Induced Measurement Artifacts

In contrast to physical regeneration, the dataset exhibits distinct, massive outliers where recorded capacity deviates erratically (e.g., instantaneous spikes exceeding 1.7 Ahr in late-stage cycling—green ellipse). The most likely conclusion is that these are documented measurement artifacts generated by the testing hardware, integration failures during the Coulomb counting process, or even manufacturing defects (owing to distinct capacity values at BOL).

Singular Value Decomposition (SVD) is highly sensitive to massive scalar outliers, which can artificially skew the dominant temporal principal components (the V matrix) away from the true aging trajectory. Therefore, prior to matrix decomposition, a median-filtering algorithm must be applied across the cycle life. This preprocessing step ensures the removal of artificial hardware spikes while preserving the genuine, underlying electrochemical degradation curve necessary for an accurate subspace representation.

Figure 3-5: Compares the degradation of battery capacity (Ahr) per cycle between isothermal battery groups operating at different temperatures.



3.1.3. Data Discrepancies and Assumed Confounding Variables

A review of the raw transient telemetry (CSVs 01259 and 01263) for the dynamic profile cell B0040 revealed severe confounding variables in the experimental execution. Coinciding with

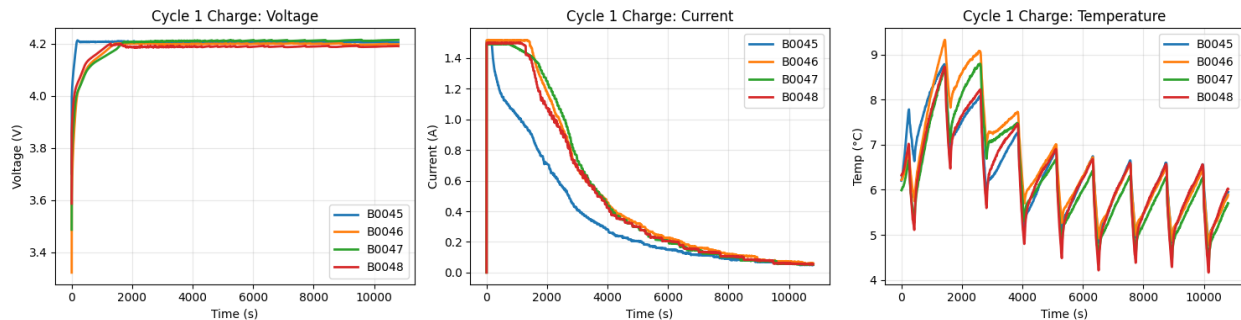
the ambient temperature shift from 24°C to 44°C, the commanded discharge load was simultaneously reduced from 4.0A to 1.0A, extending the discharge duration from ~1,380s to ~6,375s. This simultaneous alteration of thermal and electrical parameters creates inextricably linked non-linearities in the dataset. Consequently, extracting pure thermal degradation coefficients for SVD requires excluding these dynamic profiles entirely, validating the use of the strictly controlled isothermal datasets (e.g., the 1A / 24°C group).

The dataset overview defines the EOL threshold as a 20% capacity fade (1.6 Ahr). However, the empirical data shows cells frequently cycling down to ~50% fade. It is highly likely that the automated cycling scripts were set to a fixed number of cycles rather than a strict capacity cutoff, or that researchers deliberately allowed the tests to continue to gather extended degradation data.

3.1.4. Environmental Chamber Thermal Noise

In examining the transient temperature curves (Figure 3-1) for the low-temperature datasets (e.g., the 4°C tests for B0047), high-frequency oscillating spikes are present throughout the entire discharge phase. While the specific volume of the NASA environmental testing hardware is not documented, it can be reasonably assumed that these sawtooth fluctuations are environmental artifacts caused by the test chamber's cooling compressor.

Figure 3-6: Compares the variation in voltage, current and temperature profiles during the first charging cycle of different batteries in a group.



3.2. Baseline Dataset Selection & Operational Protocol

NASA Dataset (Li-Ion Battery)

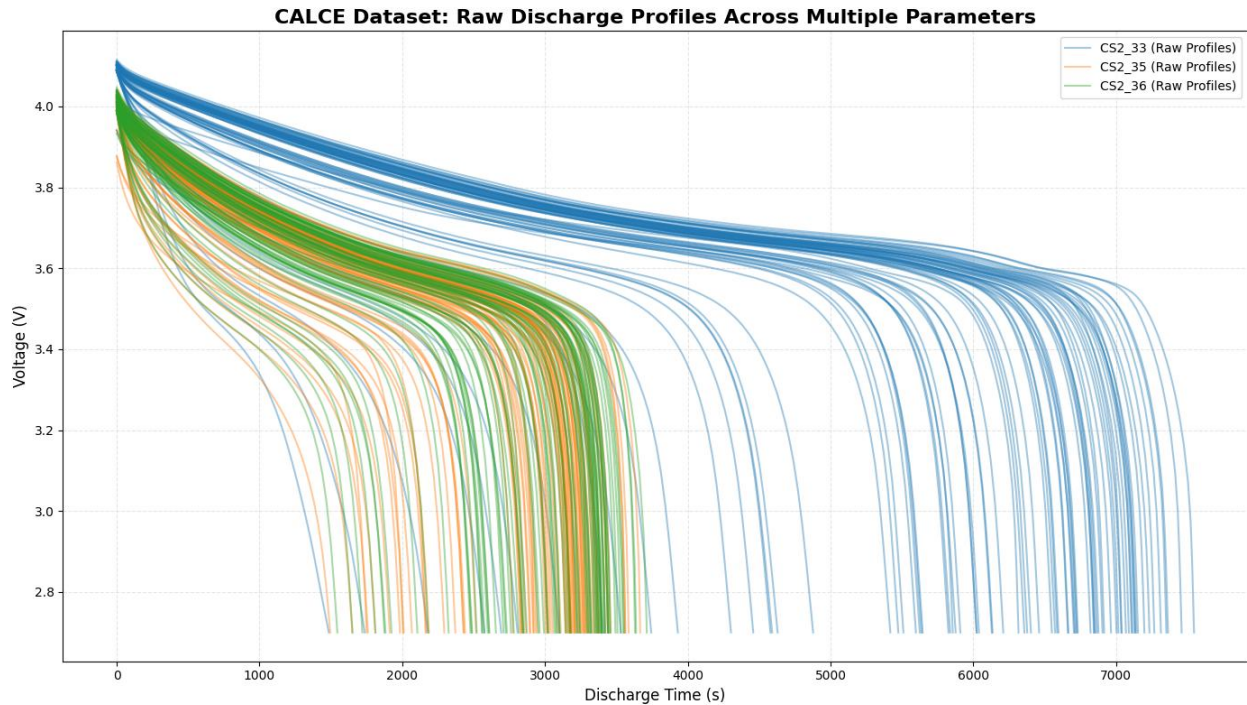
To establish a mathematically stable foundation for the Singular Value Decomposition (SVD) model, the core analysis is restricted to a strictly controlled, isothermal subset: Batteries B0005, B0006, B0007, and B0018.

This group provides the pristine degradation data required for modeling because it was subjected to continuous cycling under a constant ambient temperature of 24°C without dynamic parameter shifts. The operational profiles are defined by the following quantifiable metrics:

- **Charging Protocol:** A standard Constant Current-Constant Voltage (CC-CV) mode. Cells were driven at a 1.5A constant current until reaching a 4.2V ceiling, followed by a constant voltage taper until the current dropped to 20mA.
- **Discharging Protocol:** A strict, continuous 2.0A constant current load applied until the cells hit predefined cutoff voltages (2.7V for B0005, 2.5V for B0006/B0018, and 2.2V for B0007).
- **Impedance Tracking:** Intermittent Electrochemical Impedance Spectroscopy (EIS) frequency sweeps (0.1Hz to 5kHz) to monitor internal parameter shifts during aging.
- **End-of-Life (EOL) Criterion:** Experiments were systematically terminated when cells hit a 30% capacity fade, dropping from a nominal 2.0 Ahr to the 1.4 Ahr failure threshold.

CALCE Dataset (CS2 Battery)

Figure 3-7: Raw discharge profiles from the CALCE CS2 dataset.



3.2.1. Matrix Construction and Normalization

To facilitate the Singular Value Decomposition ($X = U\Sigma V^T$), the asynchronous discharge telemetry must be mapped onto a uniform spatial-temporal grid. Due to the monotonic decrease in capacity over the cell's life, subsequent discharge cycles contain progressively fewer data points.

To resolve this dimensional mismatch, the time domain for each valid discharge cycle was normalized from $t_0 = 0$ to $t_{end} = 1$. A linear interpolation function was then applied to project the raw measured voltage onto a rigid 3000-point normalized grid.

This transformation yields the primary data matrix $X \in \mathbb{R}^{m \times n}$, where $m = 100$ represents the spatial voltage distribution along the normalized discharge curve, and n represents the chronological cycle history. This formulation allows the SVD algorithm to cleanly separate the fundamental electrochemical discharge shape (U) from the transient degradation scaling (V).

$$X = \begin{bmatrix} V(t_1, 1) & V(t_1, 2) & \dots & V(t_1, N) \\ V(t_2, 1) & V(t_2, 2) & \dots & V(t_2, N) \\ \vdots & \vdots & \ddots & \vdots \\ V(t_m, 1) & V(t_m, 2) & \dots & V(t_m, N) \end{bmatrix}$$

By structuring the data in this manner, the physical phenomenon of capacity fading is intrinsically captured within the rows of the matrix. As the cycle number n increases (moving left to right across the columns), the internal resistance of the cell grows, causing the voltage at any given time step t to systematically depress.

Min-Max Time Normalization –

Because Cycle 1 might take 3000 seconds and Cycle 160 might take 1500 seconds, we map all time arrays to a standard $[0,1]$ domain.

$$t_{norm} = \frac{t - t_{min}}{t_{max} - t_{min}}$$

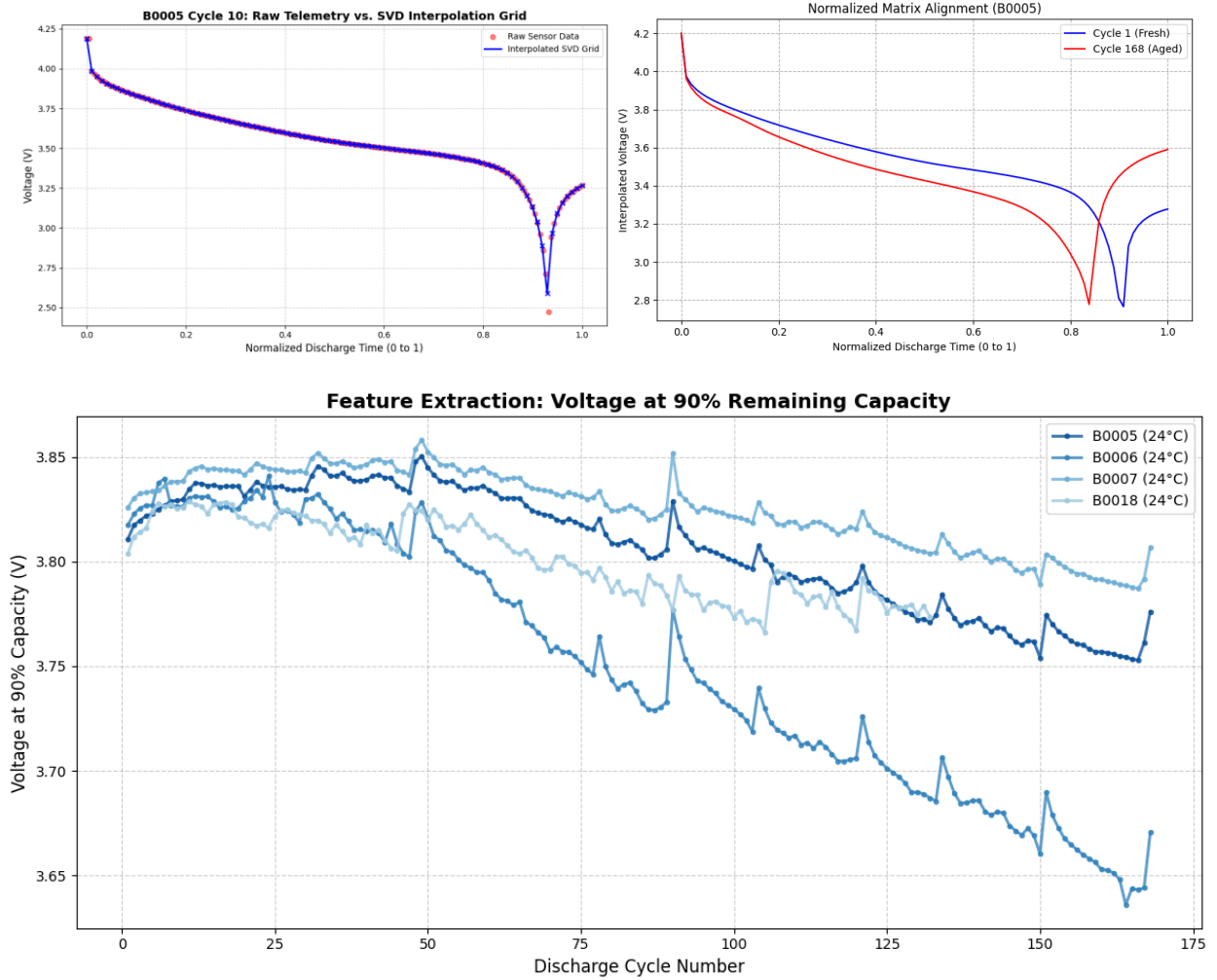
(Where t_{norm} is the new unified time, t is the raw timestamp, and t_{max} is the final second of that specific discharge cycle).

Linear Interpolation –

Once the time is normalized to a 0 to 1 scale, the code generates a rigid 3000-point grid. It calculates the exact voltage at these new artificial grid points by drawing a straight line between the two closest raw data points (V_0, t_0 and V_1, t_1).

$$V(t_{norm}) = V_0 + (t_{norm} - t_0) \frac{V_1 - V_0}{t_1 - t_0}$$

Figure 3-8: (a) SVD reconstructed for cycle, (b) Discharge cycles at BOL and EOL, (c) Voltage modeled for all cycles with 90% remaining capacity.



Alternative SVD Preprocessing Techniques:

Linear interpolation is the simplest method, but there are several other scientifically valid ways to align this dataset.

- **Dynamic Time Warping (DTW):** Instead of forcing a rigid linear grid, DTW is an algorithm that stretches and compresses two time-series curves until their “shapes” perfectly align. It is highly effective for aligning battery curves that have slightly different voltage drop-off rates (Berndt et al., n.d.)
- **Cubic Spline Resampling:** Instead of the linear formula above, this method fits a 3rd-degree polynomial curve through the raw voltage data points. It is often preferred in fluid dynamics and battery modeling because it preserves the smooth, continuous derivatives of the physical curve (Brunton & Kutz, 2022)
- **Voltage Padding (Threshold Alignment):** Instead of normalizing time, some researchers align the data purely by voltage. They take the data strictly between 4.2V

and 2.7V. If a cycle ended prematurely at 2.8V (due to hardware error), the algorithm simply “pads” the missing tail with a steady-state value or an average drop-off rate to ensure matrix dimensions match (*Battery Data Set, NASA Ames Prognostics Data Repository - Google Scholar*, n.d.)

3.2.2. Inverse Transformation: From Latent Space to Physical Reality

While the Singular Value Decomposition ($X = U\Sigma V^T$) operates on a normalized mathematical subspace, analyzing the physical behavior of the battery requires reconstructing the data back into its native units (Seconds and Volts).

This requires two distinct mathematical inverse transformations: one for the temporal axis (Time), and one for the magnitude axis (Voltage).

To recover the physical time in seconds for any specific cycle n , the normalized grid t_{norm} must be multiplied by that cycle’s original absolute duration. Assuming the discharge starts at $t = 0$, the inverse transformation is:

$$t_{physical,n} = t_{norm} \times \Delta t_n$$

(Where Δt_n is the total elapsed time of cycle n recorded in the raw telemetry).

When observing the first spatial mode (U_1), the Y-axis does not represent Volts. The matrix U consists of unit vectors (their mathematical length is exactly 1).

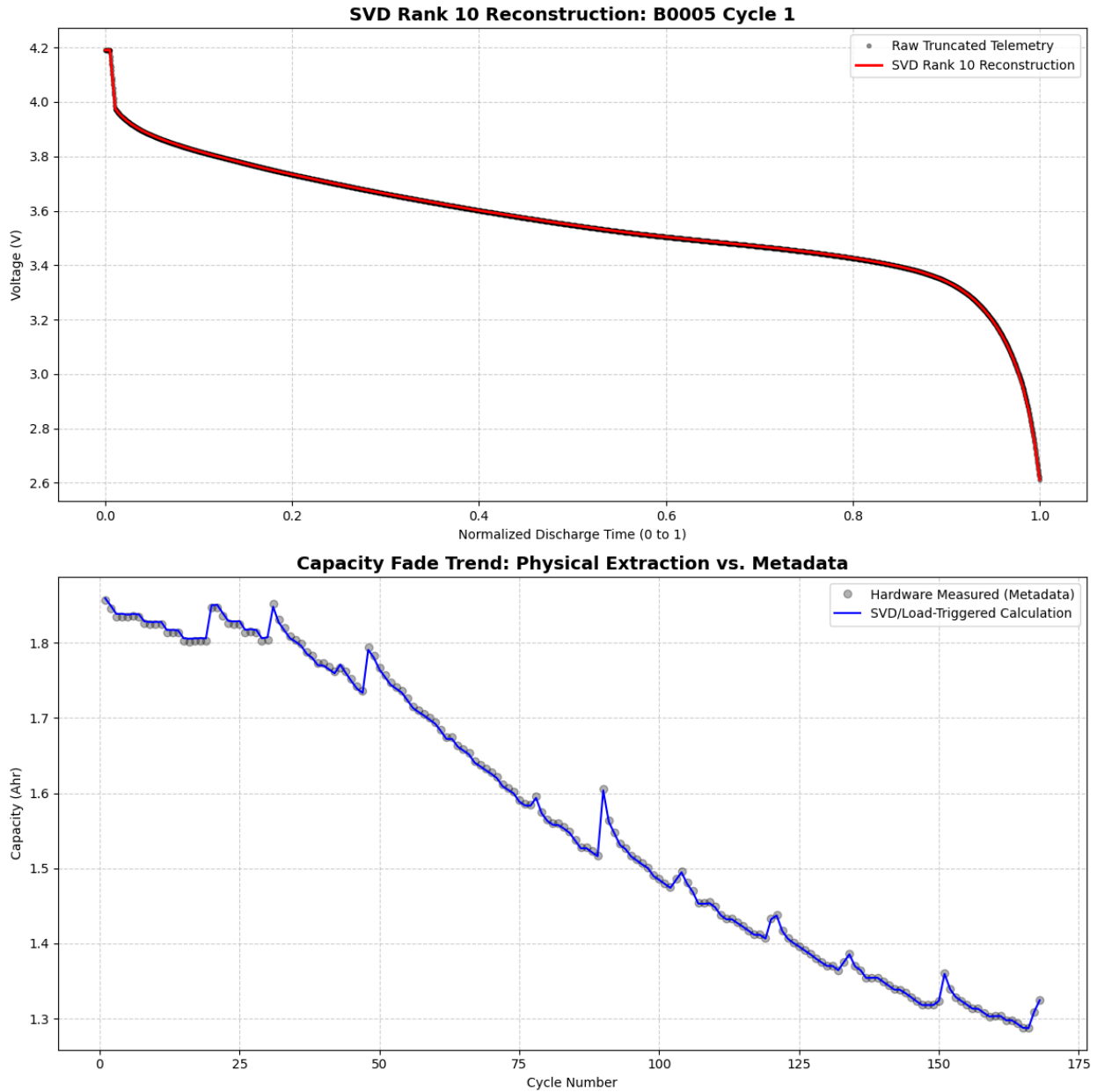
To recover the true physical voltage amplitude for a specific cycle n , the pure shape (U) must be scaled by both the systemic energy of that mode (Σ) and the cycle’s specific aging multiplier (V^T). The physical voltage curve for cycle n reconstructed using rank- r approximation is:

$$Voltage_{physical,n} = \sum_{i=1}^r (U_i \times \Sigma_{i,i} \times V_{i,n}^T)$$

4. Singular Value Decomposition

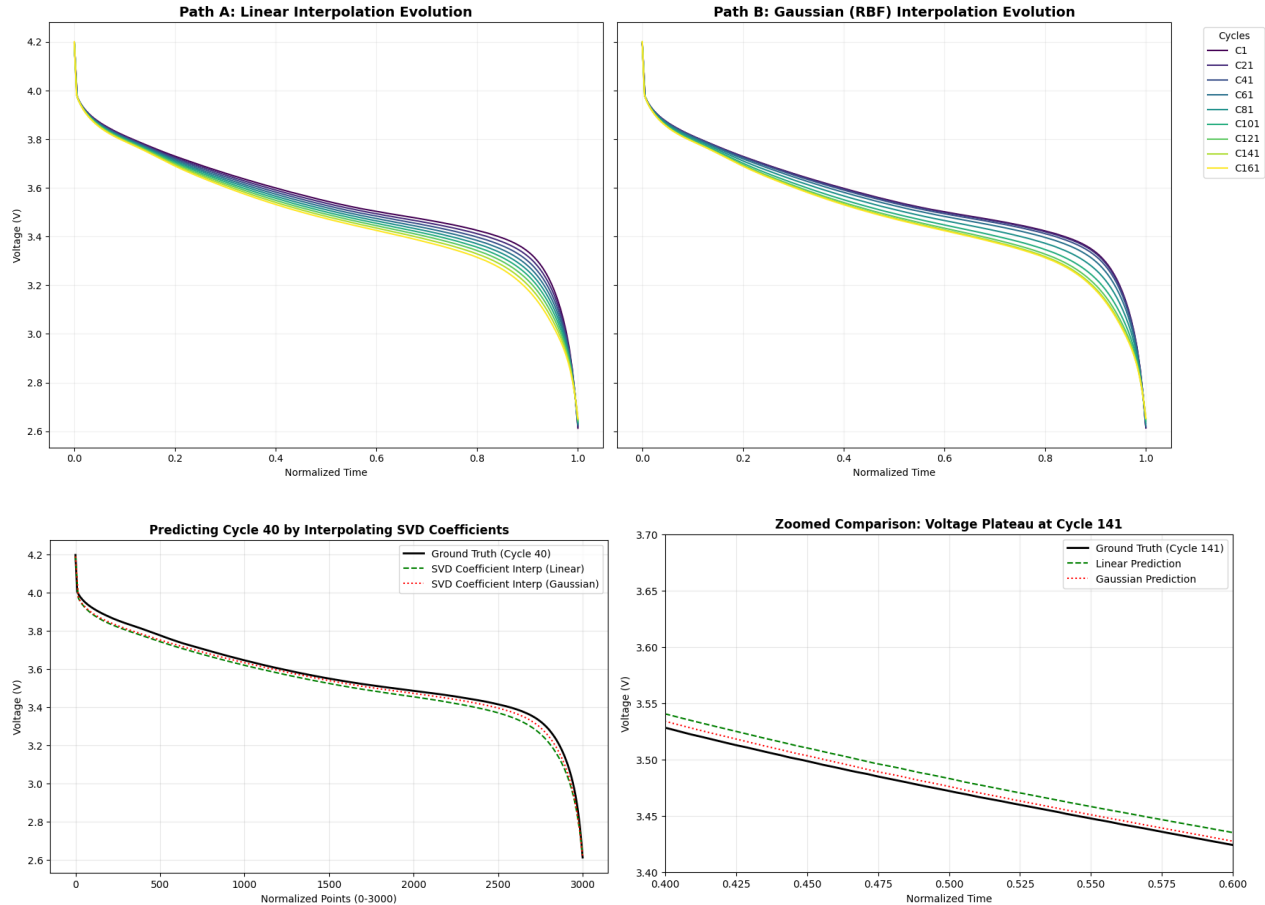
From the NASA dataset, we constructed the SVD for B0005 battery. Rank 10 (Figure 4-1) showed a very close alignment to the data. We plotted comparison plots for the discharge cycle and capacity fade. Error was below 0.02%. **We use discharge cycle as the model parameter.** We apply SVD to both the extreme cycles (1 and 160) and use linear and gaussian interpolation to predict the discharge profile of cycle 41 and 141.

Figure 4-1: (a) SVD rank 10 reconstruction, (b) Capacity fade profile reconstruction using SVD (perfect match).



The gaussian and linear interpolation comparison plots (Figure 4-2) have been presented below. While the linear interpolation has even spacing, the gaussian provides a more realistic depiction of trends during different cycles. The gaussian is much closer to ground-truth with an error of 0.01906 compared to 0.03497 of the linear interpolation (cycle 40 pred).

Figure 4-2: (a) Comparison b/w gaussian and linear discharge profile for multiple cycles, (b) discharge cycle 40, (c) discharge cycle 141

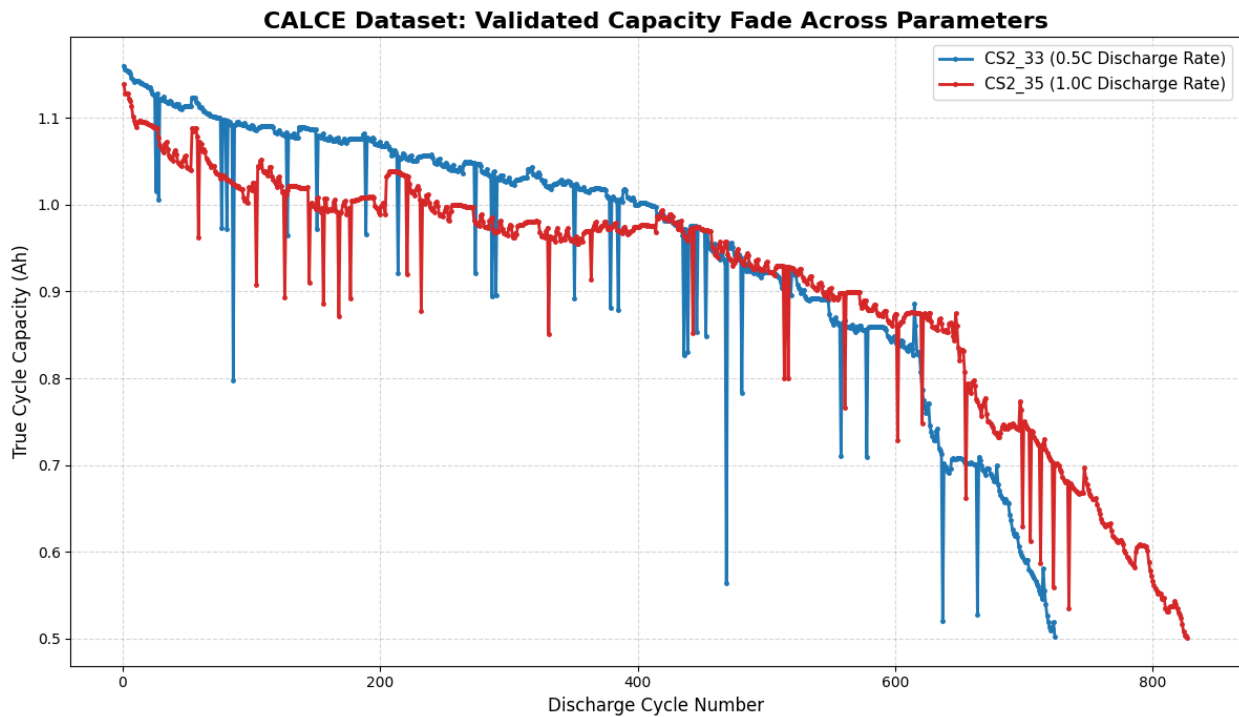
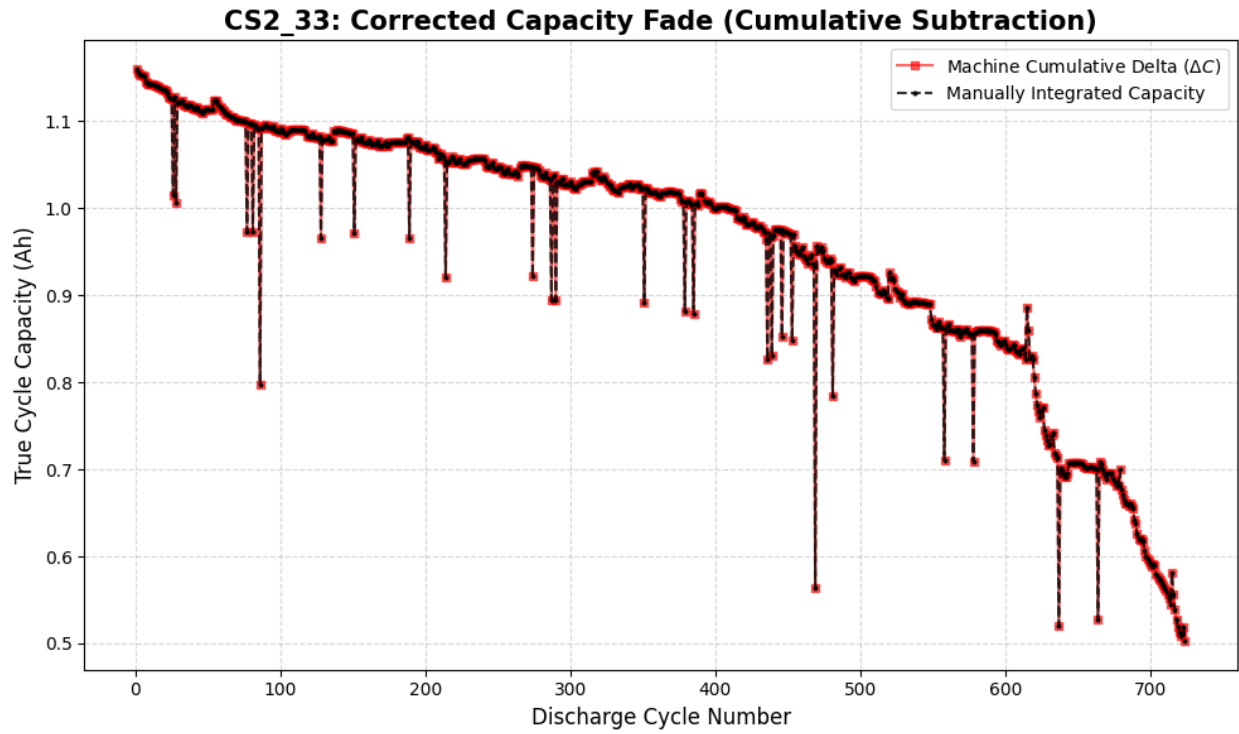


For cycle 141, the error is greater for linear interpolation (0.015) compared to gaussian (0.00789).

For the CALCE data, we applied the same methodology, by isolating the discharge cycles for the CS2_33 battery. **We model around the constant current input in this case.** We derive the capacity fade plot (Figure 4-3) from manual calculations and compare with machine data; it is a good match. There are anomalies, which are singular data points that don't follow the trend of decreasing capacity with cycle number. These can be de-noised. We compare the capacity fade of CS2_33 which has 0.5C discharge battery with CS2_35 (the 1.0C discharge battery) and plot it. Based on the plot we cannot construct a multi-parameter model as there is no direct correlation between 0.5C and 1C capacity fade plots.

NOTE: 0.5C and 1C represent constant current applied during charging/discharging phase.

Figure 4-3: Capacity fade profile for CS2_33 battery.



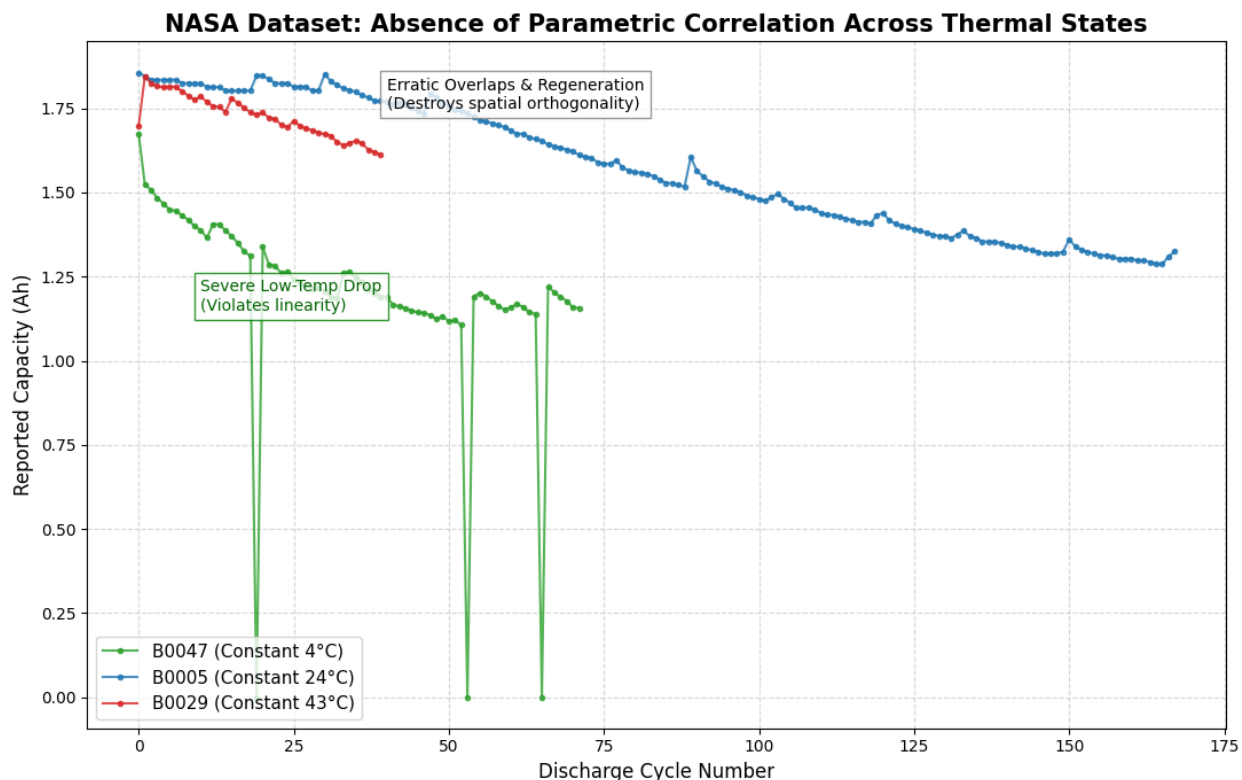
4.1. Limitations of Empirical Multi-Parameter Modeling

While spatial manifold interpolation is highly effective for deterministic, physics-based simulations (such as parameterized fluid dynamics), its application to empirical battery

degradation datasets presents severe limitations. An analysis of the CALCE CS2 capacity fade profiles across varying discharge rates (0.5C versus 1.0C) reveals high intra-group variance and non-monotonic degradation paths. Because the raw data does not exhibit a clearly separated, proportional physical correlation to the applied current parameter, traditional linear or geometric interpolation (e.g., predicting a 0.75C state) is mathematically invalid.

This lack of clean parametric correlation is not unique to the CALCE dataset. An analysis of the NASA Prognostics Center battery dataset under varying thermal conditions (e.g., constant 24°C versus elevated 43°C ambient temperatures) yields similarly erratic overlaps. Due to localized capacity regeneration phenomena, unpredictable SEI layer fracturing, and sensor noise, batteries at elevated temperatures frequently exhibit degradation curves that intersect or temporarily outperform their room-temperature counterparts. Consequently, applying standard proper orthogonal decomposition (POD) across these parameter spaces produces smeared spatial bases that fail to accurately capture the underlying electrochemistry.

Figure 4-4: Temperature condition for NASA batteries (uncorrelated capacity fade plots)



4.2. Advanced Scientific Machine Learning (SciML) and Hybrid Approaches

To circumvent the limitations of linear parametric interpolation, current state-of-the-art state-of-health (SoH) estimation has pivoted toward Scientific Machine Learning (SciML) and hybrid data-driven architectures. Rather than relying on rigid geometric manifolds, researchers utilize non-linear surrogate models to map complex parameter spaces.

Methods such as autoencoders and support vector quantile regression are increasingly deployed to predict the SOH through Health Indicators, that are formulated in Spearman Analysis (Sun et al., 2024). Hybrid architectures frequently pair Reduced-Order Models with Gaussian process regression (GPR) or recurrent networks (LSTMs) to capture the highly stochastic, non-linear dynamics of capacity regeneration that traditional linear algebra fails to resolve (Jia et al., 2020).

Non-data-driven models fundamentally rely on the physical, electrical, and chemical principles governing the cell rather than historical degradation datasets. The most rigorous of these, full-order electrochemical models, provide the major advantage of accurately depicting internal dynamic behaviors, including lithium concentration in solid and electrolyte phases, diffusion kinetics, and electric potential distributions. However, their deployment in real-time, onboard Electric Vehicle (EV) applications is severely restricted. These models require solving computationally intensive sets of partial differential equations (PDEs), and dynamically determining the unobservable chemical composition parameters of an aging battery is nearly impossible outside a laboratory setting. Consequently, practical battery management systems often compromise by utilizing simplified physics-based approaches, such as Equivalent Circuit Models (ECMs) or Single Particle Models (SPMs), which significantly reduce the computational burden while still anchoring SOH and RUL estimations in fundamental battery physics (Elmahallawy et al., 2022; Hu et al., 2014; Saha et al., 2009).

5. Conclusion

This project set out to enhance Electric Vehicle (EV) battery enclosure safety by modeling the Remaining Useful Life (RUL) of lithium-ion cells under varying thermal and operational loads. Initial efforts focused on developing a multi-parameter Reduced-Order Model (ROM) utilizing spatial manifold interpolation to predict degradation at untested continuous states. However, extensive analysis of the CALCE and NASA datasets revealed that empirical battery degradation is profoundly non-monotonic. Erratic capacity regeneration, low-temperature voltage drops, and intersecting degradation paths across different thermal parameters mathematically precluded the use of standard geometric interpolation techniques across parameter spaces, as the data violated the requisite spatial orthogonality constraints.

Despite these multi-parameter limitations, this study successfully demonstrated the power of data-driven temporal modeling for battery prognostics. By pivoting to a single-parameter framework, Singular Value Decomposition (SVD) was effectively utilized to isolate the underlying linear dynamics of capacity fade into a truncated orthogonal basis. Furthermore, the application of Gaussian interpolation to the temporal coefficients—relying exclusively on Beginning of Life (BOL) and End of Life (EOL) boundary profiles—yielded a highly accurate predictive surrogate model. This framework significantly outperformed standard linear interpolation, effectively halving the predictive error (achieving 0.00789 versus 0.015 at cycle 141) and proving its efficacy for reconstructing intermediate discharge cycles with minimal experimental training data.

Ultimately, the findings of this project highlight a definitive boundary between the capabilities of traditional linear algebra-based ROMs and the highly non-linear, chaotic reality of battery electrochemistry. To fully realize a multi-parameter prognostic model capable of simulating unmapped thermal boundary conditions, future research must transition toward advanced Scientific Machine Learning (SciML) architectures. Frameworks such as Deep Operator Networks (DeepONets) or hybrid generative models are uniquely positioned to map complex, noisy parameter spaces, building upon the foundational feature extraction of SVD to ultimately bridge the gap between discrete experimental testing and comprehensive thermal management design.

6. References

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